TASK 3: AIR PRESSURE BUILD-UP, LEAK AND WARNING-DEVICE TESTS

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 4 | v1.1

TIME	EVIDENCE	PROGRAMME POINT
90 minutes (adapt to workplace conditions)	**Practical | Knowledge | Documentation**	**Weeks 43-58 | Months 12-15**

Your task

Test air-brake system pressure build-up, governor operation, warning devices and leakage on a heavy vehicle. Compare results with approved data and report any safety-critical defect.

What you must do

1. Review the pneumatic diagram, test method and vehicle-specific precautions.
2. Secure the vehicle and connect calibrated pressure/test equipment safely.
3. Record pressure build-up, cut-in/cut-out and low-pressure warning operation.
4. Carry out specified static/applied leakage tests without entering danger zones.
5. Interpret results, restore the system and complete a safety-critical report.

What good evidence looks like

SAFE CONTROL

Chock the vehicle and keep clear of actuators, suspension and moving linkages; stored air must be exhausted by the approved method before disassembly.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle with air brakes
- Pneumatic diagram and test specifications
- Calibrated pressure gauges and timer
- Chocks, PPE and defect record

SOW ALIGNMENT

Block 4 - Air Braking, Pneumatic and Hydraulic Systems

KSB evidence focus: K1, S1, S3; reinforced K3-K6, K10, S4-S10, B2-B3

Task reference: HV_L3_Block_4_Task_3_Air_Pressure_Build_Up_Leak_and_Warning_Device_Tests

TASK 3: AIR PRESSURE BUILD-UP, LEAK AND WARNING-DEVICE TESTS

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Initial / final pressure		Build-up time	
Governor cut-in		Governor cut-out	
Low-pressure warning		Static / applied leakage	
Specified limits		Result / action	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why must the vehicle be secured independently of air pressure?

2. What does excessive build-up time indicate?

Knowledge check - continued

3. Explain the purpose of low-pressure warning devices.

4. How can you locate leakage without entering a crush zone?

5. What action is required for a result outside the safety limit?

Assessor/Workplace Mentor observation

- [ ] Secures the vehicle and controls stored pneumatic energy.
- [ ] Uses the correct diagram, gauges and test conditions.
- [ ] Records pressures, times and leakage with units.
- [ ] Compares results with approved limits.
- [ ] Escalates defects and restores the system safely.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- [ ] PASS
- [ ] REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.